

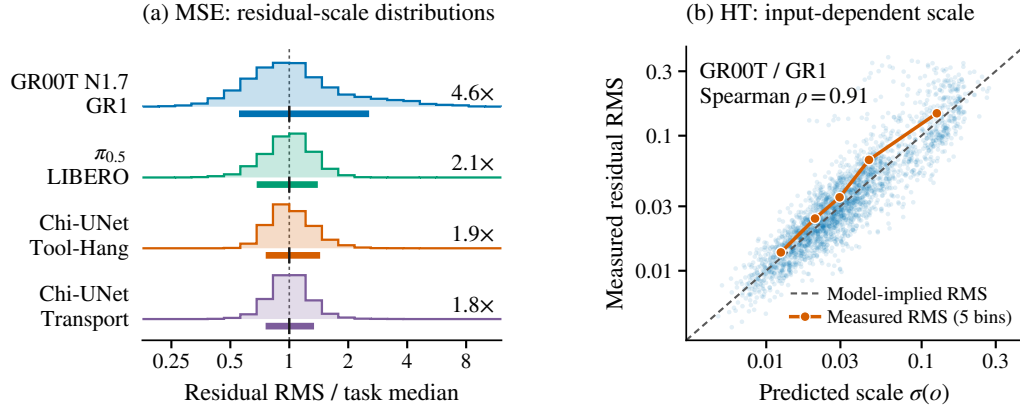


Figure 1: **Residual scales vary across demonstration inputs.** (a) MSE action-chunk residual RMS distributions, normalized by the median within each task and aggregated with equal task weights. Histograms use log-spaced bins and equal peak heights; bars mark the 10th–90th percentiles, with their ratios annotated. (b) In a separate GR1 HT probe, predicted scale tracks measured residual RMS ($\rho = 0.91$). Dots are inputs; connected markers pool five groups defined by predicted σ ; the dashed line is the model-implied RMS. All measurements use training demonstrations and continuous action channels; binary grippers are excluded and GR1 hand joints are retained.

RESIDUAL-SCALE DIAGNOSTIC

Residual magnitudes span multiple scales across demonstration inputs (Figure 1). After removing differences in median scale between tasks, the 10th–90th percentile range of MSE residual RMS spans $4.6\times$ on GR1, $2.1\times$ on LIBERO, $1.9\times$ on Tool-Hang, and $1.8\times$ on Transport. In a separate GR1 HT probe, the learned input-dependent scale is strongly associated with the measured residual RMS (Spearman $\rho = 0.91$). These fitting diagnostics motivate an input-dependent scale in the regression objective.

Scope of this diagnostic. The measurements describe fitting residuals, not isolated demonstration noise. The two panels use different sampled inputs and are not a paired MSE–HT performance comparison. The observed spread does not establish discrete Gaussian components, and the HT panel is a training-set check, not held-out calibration.